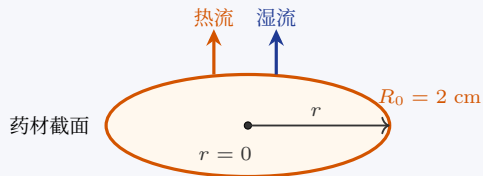


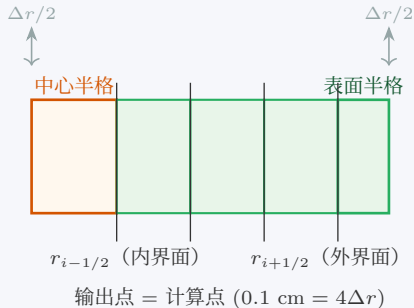
(a) 圆柱截面与边界条件

$$-k\partial_r T = h(T_R - T_\infty), \quad -D\partial_r C = k_m(C_R - C_\infty)$$



$r = 0$: 对称条件 $\partial_r T = \partial_r C = 0$

(b) 径向控制体 ($N = 80, \Delta r = 0.25 \text{ mm}$)



(c) 时空离散与推进

